The arithmetic root of degree n

Undoing a power: what the n-th root means, when it exists, and the four properties that let you work with it.

LESSON 28–29 | 2 × 40 MIN | ALGEBRA 8, §8 | STAGE 9 · 1.1

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define the arithmetic root of degree n and say when it exists.
- Use the four properties of roots to simplify.
- Take a factor out of a root and put one back in.
- Rationalise a simple denominator.

TEXTBOOK

National	§8, pp. 39–41
Cambridge	Learner's Book pp. 10–14
Workbook	Workbook 1.1

01 Explanation

The definition

DEFINITION

For $a \geq 0$ and a natural number $n \geq 2$, the **arithmetic root of degree n** of a is the **non-negative** number whose n-th power is a:

$$\sqrt[n]{a} = b \quad \text{means} \quad b \geq 0 \text{ and } b^n = a$$

So $\sqrt{16} = 4$ — and only 4. The number -4 also squares to 16, but the *arithmetic* root is the non-negative one by definition.

WHEN DOES IT EXIST?

For **even** n the radicand must satisfy $a \geq 0$ — you cannot take $\sqrt{-9}$. For **odd** n any a is allowed: $\sqrt[3]{-8} = -2$.

The four properties

For $a \geq 0$, $b \geq 0$:

$$\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b} \qquad \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}},\ b > 0$$

$$(\sqrt[n]{a})^k = \sqrt[n]{a^k} \qquad \sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$

Read the first one right to left and you can multiply roots together; read it left to right and you can **take a factor out**:

$$\sqrt{50} = \sqrt{25 \cdot 2} = \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2}$$

THERE IS NO RULE FOR A SUM

$\sqrt{a + b}$ is **not** $\sqrt{a} + \sqrt{b}$. Check it: $\sqrt{9 + 16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$.

Rationalising the denominator

A root in the denominator is untidy. Multiply top and bottom by the same root:

$$\frac{1}{\sqrt{3}} = \frac{1 \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} = \frac{\sqrt{3}}{3}$$

This is the fundamental property of a fraction again — the same tool as in Quarter I.

$\sqrt{a} \rightarrow a^{(1/2)}$

$\sqrt[3]{a} \rightarrow a^{(1/3)}$

$\sqrt[4]{a} \rightarrow a^{(1/4)}$

$\sqrt[n]{a^m} \rightarrow a^{(m/n)}$

a root is a power — the same object, two notations

Every root can be written as a power, which is the bridge to the next lesson.

02 Worked examples

EXAMPLE 1 ***Simplify*** $\sqrt{72}$

1. $72 = 36 \cdot 2$

   Look for the largest square factor.

2. $\sqrt{72} = \sqrt{36} \cdot \sqrt{2}$

   Property 1.

3. $= 6\sqrt{2}$

ANSWER

$6\sqrt{2}$

EXAMPLE 2 ***Simplify*** $\sqrt{12} \cdot \sqrt{3}$

1. $\sqrt{12} \cdot \sqrt{3} = \sqrt{36}$

   Property 1, read right to left.

2. $= 6$

   A whole number — worth checking for every time.

ANSWER

6

EXAMPLE 3 ***Write*** $\dfrac{6}{\sqrt{2}}$ ***without a root in the denominator***

(1) $\dfrac{6 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$

Multiply top and bottom by $\sqrt{2}$.

(2) $= \dfrac{6\sqrt{2}}{2}$

$\sqrt{2} \cdot \sqrt{2} = 2$

(3) $= 3\sqrt{2}$

Cancel the 2.

ANSWER

$3\sqrt{2}$

03 Interactive model

Move the index n and watch the same number appear as a root and as a power.

A root is a power

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Root of degree n	n-darajali ildiz	Корень n-й степени
Arithmetic root	Arifmetik ildiz	Арифметический корень
Radicand (under the root)	Ildiz ostidagi ifoda	Подкоренное выражение
Index of the root	Ildiz ko‘rsatkichi	Показатель корня
Square root	Kvadrat ildiz	Квадратный корень
Cube root	Kub ildiz	Кубический корень
Irrational number	Irratsional son	Иррациональное число
Rationalise the denominator	Maxrajni irratsionallikdan qutqarish	Освободиться от иррациональности в знаменателе
Take a factor out	Ko‘paytuvchini ildiz oldiga chiqarish	Вынести множитель из-под корня
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sqrt{49}$ equals:

- 7
- ± 7
- -7
- 24.5

$\sqrt[3]{-27}$ equals:

- it does not exist
- -3
- 3
- -9

$\sqrt{8} \cdot \sqrt{2}$ equals:

- 4
- $\sqrt{10}$
- 16
- $2\sqrt{2}$

$\sqrt{9 + 16}$ equals:

- 7
- 5
- 25
- $3 + 4$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find* $\sqrt{36}$

   ANSWER 6

2. *Find* $\sqrt{81}$

   ANSWER 9

3. *Find* $\sqrt[3]{8}$

   ANSWER 2

4. *Find* $\sqrt[3]{27}$

   ANSWER 3

5. *Find* $\sqrt{100} + \sqrt{25}$

   ANSWER 15

6. *Does* $\sqrt{-4}$ *exist?*

   ANSWER No — an even root needs $a \geq 0$.

7. *Find* $\sqrt[3]{-8}$

   ANSWER -2

Medium · the standard question

7 PROBLEMS

1. *Simplify* $\sqrt{72}$

   ANSWER $6\sqrt{2}$

2. *Simplify* $\sqrt{50}$

   ANSWER $5\sqrt{2}$

3. *Simplify* $\sqrt{12} \cdot \sqrt{3}$

   ANSWER 6

4. *Simplify* $\dfrac{\sqrt{18}}{\sqrt{2}}$

   ANSWER 3

5. *Simplify* $\sqrt{45}$

   ANSWER $3\sqrt{5}$

6. *Write* $\dfrac{6}{\sqrt{2}}$ *without a root below*

   ANSWER $3\sqrt{2}$

7. *Simplify* $2\sqrt{3} + 5\sqrt{3}$

   ANSWER $7\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. *Simplify* $\sqrt{98} - \sqrt{50}$

ANSWER $7\sqrt{2} - 5\sqrt{2} = 2\sqrt{2}$

2. *Simplify* $\sqrt[4]{16}$

ANSWER 2

3. *Simplify* $(\sqrt{5})^2 + \sqrt{16}$

ANSWER $5 + 4 = 9$

4. *Write* $\dfrac{10}{\sqrt{5}}$ *without a root below*

ANSWER $2\sqrt{5}$

5. *Simplify* $\sqrt{2} \cdot \sqrt{8} - 1$

ANSWER $\sqrt{16} - 1 = 3$

6. *Show that* $\sqrt{9 + 16} \neq \sqrt{9} + \sqrt{16}$

ANSWER $5 \neq 7$

7. *Simplify* $\sqrt{a^2}$ *for* $a \geq 0$, *and for* $a < 0$

ANSWER a when $a \geq 0$; $-a$ when $a < 0$.

07 Homework

Homework — 5 problems

Algebra 8, §8, pp. 39–41.

1. *Find* $\sqrt{144}$, $\sqrt[3]{64}$ *and* $\sqrt[3]{-125}$
2. *Simplify* $\sqrt{32}$

3. *Simplify* $\sqrt{20} \cdot \sqrt{5}$

4. *Write* $\dfrac{8}{\sqrt{2}}$ *without a root in the denominator*

5. *Simplify* $3\sqrt{7} - \sqrt{7}$

Powers with a rational exponent

One definition joins roots and powers into a single system — and all the old index laws keep working.

LESSON 30–31 | 2 × 40 MIN | ALGEBRA 8, §9 | STAGE 9 · 1.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define $a^{1/n}$ and $a^{m/n}$.
- Convert freely between root notation and index notation.
- Use the index laws with fractional and negative exponents.
- Evaluate powers with a rational exponent without a calculator.

TEXTBOOK

National	§9, pp. 42–48
Cambridge	Learner's Book pp. 16–20
Workbook	Workbook 1.3

01 Explanation

The definition

DEFINITION

For $a > 0$ and natural m, n with $n \geq 2$:

$$a^{1/n} = \sqrt[n]{a} \quad \text{and} \quad a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

and, as before, $a^0 = 1$ and $a^{-k} = \dfrac{1}{a^k}$.

Why this definition and no other? Because it is the only one that keeps the index laws true. If $(a^{1/2})^2 = a^1 = a$ then $a^{1/2}$ must be the number that squares to a — that is $\sqrt{a}$.

The **denominator of the index is the root**, the **numerator is the power**. Say it out loud once and the whole topic becomes bookkeeping.

The index laws, unchanged

$$a^p \cdot a^q = a^{p+q} \quad \cdot \quad a^p : a^q = a^{p-q} \quad \cdot \quad (a^p)^q = a^{pq}$$

$$(ab)^p = a^p b^p \quad \cdot \quad \left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$$

The letters p and q are now any rational numbers, positive or negative. Nothing else changes.

Evaluating without a calculator

Take the **root first**, then the power — the numbers stay small.

$$8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$$

Going the other way, $\sqrt[3]{8^2} = \sqrt[3]{64} = 4$ — same answer, bigger arithmetic.

NEGATIVE INDEX

a^{-p} means **one over** a^p; it does not make the answer negative. $4^{-1/2} = \frac{1}{4^{1/2}} = \frac{1}{2}$.

02 Worked examples

EXAMPLE 1 ***Evaluate*** $27^{2/3}$

1. Denominator 3 → cube root; numerator 2 → square.
   Root first.
2. $\sqrt[3]{27} = 3$
3. $3^2 = 9$

ANSWER

9

EXAMPLE 2 ***Evaluate $16^{-3/4}$***

1. $16^{-3/4} = \dfrac{1}{16^{3/4}}$

   A negative index means a reciprocal.

2. $\sqrt[4]{16} = 2$

   Root first.

3. $2^3 = 8$

4. $= \dfrac{1}{8}$

ANSWER

$\dfrac{1}{8}$

EXAMPLE 3 ***Simplify $a^{1/2} \cdot a^{1/3}$***

1. $a^{1/2} \cdot a^{1/3} = a^{1/2 + 1/3}$

   Add the indices.

2. $\dfrac{1}{2} + \dfrac{1}{3} = \dfrac{5}{6}$

   Common denominator 6.

3. $= a^{5/6}$

   or $\sqrt[6]{a^5}$

ANSWER

$a^{5/6}$

03 Interactive model

Set n = 3, m = 2, a = 8 and read both notations giving 4.

Root notation and index notation

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rational exponent	Ratsional ko‘rsatkich	Рациональный показатель
Power	Daraja	Степень
Base	Asos	Основание
Negative index	Manfiy ko‘rsatkich	Отрицательный показатель
Zero index	Nol ko‘rsatkich	Нулевой показатель
Index law	Daraja xossasi	Свойство степени
Reciprocal	Teskari son	Обратное число
Equivalent form	Ekvivalent ko‘rinish	Равносильная форма
Evaluate	Qiymatini topish	Вычислить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$9^{1/2}$ equals:

- 4.5
- 3
- 81
- 18

$8^{2/3}$ equals:

- 4
- 16
- $\frac{16}{3}$
- 6

$a^{-1/2}$ equals:

- $-\sqrt{a}$
- $\frac{1}{\sqrt{a}}$
- $\sqrt{-a}$
- a^2

$a^{1/2} \cdot a^{1/2}$ equals:

- $a^{1/4}$
- a
- $2a^{1/2}$
- a^2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Write* $\sqrt{a}$ *as a power*

ANSWER $a^{1/2}$

2. *Write* $\sqrt[3]{a}$ *as a power*

ANSWER $a^{1/3}$

3. *Write* $a^{1/4}$ *as a root*

ANSWER $\sqrt[4]{a}$

4. *Evaluate* $9^{1/2}$

ANSWER 3

5. *Evaluate* $8^{1/3}$

ANSWER 2

6. *Evaluate* $16^{1/2}$

ANSWER 4

7. *Evaluate* 5^0

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. *Evaluate* $27^{2/3}$

ANSWER 9

2. *Evaluate* $16^{3/4}$

ANSWER 8

3. *Evaluate* $4^{-1/2}$

ANSWER $\frac{1}{2}$

4. *Simplify* $a^{1/2} \cdot a^{1/3}$

ANSWER $a^{5/6}$

5. *Simplify* $a^{3/4} : a^{1/4}$

ANSWER $a^{1/2}$

6. *Simplify* $(a^{2/3})^3$

ANSWER a^2

7. *Evaluate* $25^{3/2}$

ANSWER 125

Hard · stretch and reason

7 PROBLEMS

1. *Evaluate* $16^{-3/4}$

   ANSWER $\frac{1}{8}$

2. *Evaluate* $32^{2/5}$

   ANSWER 4

3. *Simplify* $(a^{1/2}b^{1/3})^6$

   ANSWER a^3b^2

4. *Simplify* $\frac{a^{5/6}}{a^{1/3}}$

   ANSWER $a^{1/2}$

5. *Evaluate* $(\frac{1}{8})^{-2/3}$

   ANSWER 4

6. *Write* $\sqrt{a} \cdot \sqrt[3]{a}$ *as one power*

   ANSWER $a^{5/6}$

7. *Evaluate* $81^{0.75}$

   ANSWER $81^{3/4} = 27$

07 Homework

Homework — 6 problems

Algebra 8, §9, pp. 42–48. Take the root first, then the power.

1. *Evaluate* $64^{1/3}$ *and* $64^{2/3}$
2. *Evaluate* $100^{1/2}$ *and* $100^{-1/2}$
3. *Simplify* $a^{2/5} \cdot a^{3/5}$

4. *Simplify* $(b^{3/4})^8$

5. *Evaluate* $125^{2/3}$

6. *Write* $\sqrt[4]{a^3}$ *as a power with a rational index*

Simplifying expressions with rational exponents

The index laws applied to letters — and the factorising tricks that work just as well with fractional powers.

LESSON 32–33 | 2 × 40 MIN | ALGEBRA 8, §10 | STAGE 9 · 2.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Simplify products and quotients of powers with rational exponents.
- Take a common power out as a factor.
- Recognise a difference of squares in $a - b = (\sqrt{a})^2 - (\sqrt{b})^2$.
- Write an answer in root form when asked.

TEXTBOOK

National	§10, pp. 49–52
Cambridge	Learner's Book pp. 30–33
Workbook	Workbook 2.3

01 Explanation

The routine

1. Turn every root into a power. Fractions of indices are easier than nested radicals.
2. Apply the index laws.
3. Collect, factorise or cancel.
4. Convert back to root form only if the question asks for it.

$$\frac{\sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[6]{a}} = \frac{a^{1/2} \cdot a^{1/3}}{a^{1/6}} = a^{1/2 + 1/3 - 1/6} = a^{2/3}$$

Taking out a common power

The smallest index is the one to take out — exactly as with whole-number powers:

$$a^{3/2} + a^{1/2} = a^{1/2}(a + 1)$$

Check by multiplying back: $a^{1/2} \cdot a = a^{3/2}$ ✓ and $a^{1/2} \cdot 1 = a^{1/2}$ ✓.

The difference-of-squares trick

Since $a = (\sqrt{a})^2$ for $a \geq 0$, any difference of two non-negative numbers is a difference of squares:

$$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$$

That is the single most useful move in this section. It cancels fractions such as

$$\frac{a - b}{\sqrt{a} - \sqrt{b}} = \sqrt{a} + \sqrt{b}$$

and it also rationalises a two-term denominator, by multiplying by the **conjugate**:

$$\frac{1}{\sqrt{5} - \sqrt{3}} = \frac{\sqrt{5} + \sqrt{3}}{(\sqrt{5})^2 - (\sqrt{3})^2} = \frac{\sqrt{5} + \sqrt{3}}{2}$$

02 Worked examples

EXAMPLE 1 *Simplify* $\dfrac{a^{3/4} \cdot a^{1/2}}{a^{1/4}}$

1. $\dfrac{3}{4} + \dfrac{1}{2} = \dfrac{5}{4}$

   Multiply on top: add the indices.

2. $\dfrac{5}{4} - \dfrac{1}{4} = 1$

   Divide: subtract the index.

3. $= a$

ANSWER

a

EXAMPLE 2 *Factorise* $a^{3/2} - a^{1/2}$

1. *smallest index is* $\dfrac{1}{2}$

   That is the common factor.

2. $a^{1/2}(a^{1} - 1)$

   Subtract ½ from each index.

3. $= a^{1/2}(a - 1)$

   or $\sqrt{a}(a - 1)$

ANSWER

$\sqrt{a}(a - 1)$

EXAMPLE 3 **Simplify** $\dfrac{x-9}{\sqrt{x}-3}$, $x \geq 0, x \neq 9$

1. $x = (\sqrt{x})^2$ and $9 = 3^2$

   A difference of squares in disguise.

2. $x - 9 = (\sqrt{x} - 3)(\sqrt{x} + 3)$

3. $= \sqrt{x} + 3$

   Cancel the bracket.

ANSWER

$\sqrt{x} + 3$

03 Interactive model

Use the model to check any answer numerically: put a value in and compare both forms.

Check an index answer numerically

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Simplify	Soddalashtirish	Упростить
Common power	Umumiy daraja	Общая степень
Substitution	Almashtirish	Замена
Difference of squares	Kvadratlar ayirmasi	Разность квадратов
Factor out	Qavsdan chiqarish	Вынести за скобку
Root form	Ildiz ko‘rinishi	Форма с корнем
Index form	Daraja ko‘rinishi	Степенная форма
Expression	Ifoda	Выражение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$a^{1/2} \cdot a^{1/2} \cdot a^{1/2}$ equals:

- $a^{1/8}$
- $a^{3/2}$
- $3a^{1/2}$
- $a^{1/6}$

$a^{3/2} + a^{1/2}$ factorises to:

- $a^{1/2}(a + 1)$
- $a(a^{1/2} + 1)$
- $a^2(a + 1)$
- it does not factorise

$\frac{x - 4}{\sqrt{x} - 2}$ equals:

- $\sqrt{x} - 2$
- $\sqrt{x} + 2$
- $x + 2$
- 2

To rationalise $\frac{1}{\sqrt{5} - \sqrt{3}}$ multiply by:

- $\sqrt{5}$
- $\sqrt{5} + \sqrt{3}$
- $\sqrt{5} - \sqrt{3}$
- 2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Simplify* $a^{1/2} \cdot a^{1/2}$

   ANSWER a

2. *Simplify* $a^{3/4} : a^{1/4}$

   ANSWER $a^{1/2}$

3. *Simplify* $(a^{1/3})^3$

   ANSWER a

4. *Simplify* $a^{1/2} \cdot a$

   ANSWER $a^{3/2}$

5. *Write* $\sqrt{a} \cdot \sqrt{a}$ *more simply*

   ANSWER a

6. *Simplify* $(4a^2)^{1/2}, a \geq 0$

   ANSWER $2a$

7. *Simplify* $a^{1/2} + 3a^{1/2}$

   ANSWER $4a^{1/2}$

Medium · the standard question

7 PROBLEMS

1. *Simplify* $\dfrac{a^{3/4} \cdot a^{1/2}}{a^{1/4}}$

   ANSWER a

2. *Factorise* $a^{3/2} - a^{1/2}$

   ANSWER $\sqrt{a}(a - 1)$

3. *Simplify* $\dfrac{x - 9}{\sqrt{x} - 3}$

   ANSWER $\sqrt{x} + 3$

4. *Simplify* $(a^{1/2}b^{1/2})^4$

   ANSWER a^2b^2

5. *Simplify* $\sqrt{a} \cdot \sqrt[3]{a}$

   ANSWER $a^{5/6}$

6. *Simplify* $\dfrac{a}{\sqrt{a}}$, $a > 0$

   ANSWER $\sqrt{a}$

7. *Factorise* $a + a^{1/2}$

   ANSWER $a^{1/2}(a^{1/2} + 1)$

Hard · stretch and reason

7 PROBLEMS

1. *Simplify* $\dfrac{\sqrt{a} \cdot \sqrt[3]{a}}{\sqrt[6]{a}}$

ANSWER $a^{2/3}$

2. *Simplify* $\dfrac{x - 1}{\sqrt{x} - 1}$, $x \geq 0, x \neq 1$

ANSWER $\sqrt{x} + 1$

3. *Rationalise* $\dfrac{1}{\sqrt{5} - \sqrt{3}}$

ANSWER $\dfrac{\sqrt{5} + \sqrt{3}}{2}$

4. *Simplify* $(a^{2/3} - b^{2/3}) : (a^{1/3} - b^{1/3})$

ANSWER $a^{1/3} + b^{1/3}$

5. *Simplify* $\dfrac{a^{3/2} - a^{1/2}}{a - 1}$, $a > 0, a \neq 1$

ANSWER $a^{1/2}$

6. *Evaluate* $(\sqrt{7} + \sqrt{3})(\sqrt{7} - \sqrt{3})$

ANSWER $7 - 3 = 4$

7. *Simplify* $(a^{1/2} + 1)^2 - (a^{1/2} - 1)^2$

ANSWER $4a^{1/2}$

07 Homework

Homework — 5 problems

Algebra 8, §10, pp. 49–52.

1. *Simplify* $a^{2/3} \cdot a^{1/3}$
2. *Simplify* $\dfrac{b^{7/4}}{b^{3/4}}$
3. *Factorise* $x^{3/2} + x^{1/2}$
4. *Simplify* $\dfrac{x - 16}{\sqrt{x} - 4}$
5. *Rationalise* $\dfrac{2}{\sqrt{7} - \sqrt{5}}$

Standard form, and fractions as decimals

The national plan's "practical and interdisciplinary" lessons, carrying two Cambridge Stage 9 topics the programme does not otherwise reach.

LESSON 34–35 | 2 × 40 MIN | ALGEBRA 8, CHAPTER I PRACTICAL | STAGE 9 · 1.2, 8.1 [CAMBRIDGE INSERT]

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Write large and small numbers in standard form $a \times 10^n$, $1 \le a < 10$.
- Multiply and divide numbers written in standard form.
- Convert a fraction to a terminating or recurring decimal, and back.
- Tell a rational number from an irrational one by its decimal.

TEXTBOOK

National	pp. 62–67
Cambridge	Learner's Book pp. 14–16, 162–166
Workbook	Workbook 1.2, 8.1

01 Explanation

Standard form

DEFINITION

A number is in **standard form** when it is written as $a \times 10^n$ with $1 \le a < 10$ and n a whole number.

- $4\,500\,000 = 4.5 \times 10^6$ — the point moves 6 places left, so n is positive.
- $0.00032 = 3.2 \times 10^{-4}$ — the point moves 4 places right, so n is negative.

To calculate, handle the numbers and the powers of ten separately:

$$(3 \times 10^5) \cdot (2 \times 10^3) = 6 \times 10^8$$

$$(6 \times 10^5) : (3 \times 10^2) = 2 \times 10^3$$

TIDY THE ANSWER

40×10^4 is not in standard form. Rewrite it as 4×10^5.

Fractions as decimals

Divide the numerator by the denominator. Exactly one of two things happens:

- the division **stops** — a **terminating** decimal, e.g. $\frac{3}{8} = 0.375$;
- a block of digits **repeats for ever** — a **recurring** decimal, e.g. $\frac{1}{3} = 0.333\ldots = 0.(3)$ and $\frac{5}{11} = 0.454545\ldots = 0.(45)$.

WHICH ONE?

Cancel the fraction first. If the denominator's only prime factors are 2 and 5, the decimal terminates; otherwise it recurs. $\frac{7}{40}$ terminates ($40 = 2^3 \cdot 5$); $\frac{7}{30}$ recurs ($30 = 2 \cdot 3 \cdot 5$).

Every fraction gives a terminating or recurring decimal, and every terminating or recurring decimal comes from a fraction. A decimal that never stops *and* never repeats belongs to an **irrational** number — $\sqrt{2} = 1.41421356\ldots$, $\pi = 3.14159\ldots$.

Turning a recurring decimal back into a fraction

Multiply by a power of ten so that one whole period shifts past the point, then subtract.

$$x = 0.(7) \Longrightarrow 10x = 7.(7) \Longrightarrow 9x = 7 \Longrightarrow x = \frac{7}{9}$$

For a two-digit period multiply by 100: $x = 0.(45) \Longrightarrow 100x - x = 45 \Longrightarrow x = \frac{45}{99} = \frac{5}{11}$.

02 Worked examples

EXAMPLE 1 **Write 0.00047 and 92 000 in standard form.**

1. $0.00047 \rightarrow 4.7$

   The point moves 4 places right.

2. $= 4.7 \times 10^{-4}$

   Moving right gives a negative index.

3. $92\ 000 \rightarrow 9.2$

   The point moves 4 places left.

4. $= 9.2 \times 10^{4}$

ANSWER

4.7×10^{-4} and 9.2×10^{4}

EXAMPLE 2 **Work out $(8 \times 10^{6}) : (4 \times 10^{2})$**

1. $8 : 4 = 2$

   The numbers.

2. $10^{6} : 10^{2} = 10^{4}$

   Subtract the indices.

3. $= 2 \times 10^{4}$

   Already in standard form.

ANSWER

$2 \times 10^{4} = 20\ 000$

EXAMPLE 3 ***Write 0.(6) as a fraction***

1. $x = 0.666\ldots$

   Name the number.

2. $10x = 6.666\ldots$

   One period is one digit, so multiply by 10.

3. $10x - x = 6$

   The tails cancel exactly.

4. $9x = 6, x = \frac{6}{9} = \frac{2}{3}$

   Cancel.

ANSWER

$$\frac{2}{3}$$

03 Interactive model

Ask for an estimate in standard form before any calculation — it catches most place-value slips.

Standard form and recurring decimals

> 0.00058

1. Move the point until one non-zero digit is in front:

5.8 .

That is the value of a.

2. The point moved

4 places right .

Right $\Longrightarrow$ negative index.

3. 5.8×10^{-4}

Check: 5.8 ÷ 10 000 = 0.00058 ✓

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Standard form	Standart ko‘rinish	Стандартный вид
Power of ten	O‘nning darajasi	Степень десяти
Significant figures	Ahamiyatli raqamlar	Значащие цифры
Terminating decimal	Chekli o‘nli kasr	Конечная десятичная дробь
Recurring decimal	Davriy o‘nli kasr	Периодическая десятичная дробь
Period	Davr	Период
Rational number	Ratsional son	Рациональное число
Irrational number	Irratsional son	Иррациональное число
Convert	Aylantirish	Преобразовать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

0.00062 in standard form is:

- 6.2×10^{-4}
- 6.2×10^{4}
- 62×10^{-5}
- 0.62×10^{-3}

$(2 \times 10^{3}) \cdot (5 \times 10^{4})$ equals:

- 10×10^{7}
- 1×10^{8}
- 7×10^{7}
- 10^{12}

Which fraction gives a recurring decimal?

- $\frac{3}{8}$
- $\frac{7}{20}$
- $\frac{5}{6}$
- $\frac{9}{25}$

$0.(3)$ as a fraction is:

- $\frac{3}{10}$
- $\frac{1}{3}$
- $\frac{3}{99}$
- $\frac{1}{30}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Write 5 000 in standard form*

ANSWER 5×10^3

2. *Write 0.007 in standard form*

ANSWER 7×10^{-3}

3. *Write 3.2×10^4 as an ordinary number*

ANSWER 32 000

4. *Write 6×10^{-2} as an ordinary number*

ANSWER 0.06

5. *Write $\frac{1}{4}$ as a decimal*

ANSWER 0.25

6. *Write $\frac{1}{3}$ as a decimal*

ANSWER 0.(3)

7. *Write 0.5 as a fraction in lowest terms*

ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. *Write 0.00047 in standard form*

   ANSWER 4.7×10^{-4}

2. *Write 92 000 in standard form*

   ANSWER 9.2×10^{4}

3. *Work out* $(3 \times 10^{5}) \cdot (2 \times 10^{3})$

   ANSWER 6×10^{8}

4. *Work out* $(8 \times 10^{6}) : (4 \times 10^{2})$

   ANSWER 2×10^{4}

5. *Does* $\frac{7}{40}$ *terminate? Give the decimal.*

   ANSWER Yes — *0.175*

6. *Write* $\frac{5}{11}$ *as a decimal*

   ANSWER $0.(45)$

7. *Write 0.(6) as a fraction*

   ANSWER $\frac{2}{3}$

Hard · stretch and reason

7 PROBLEMS

1. *Work out* $(4 \times 10^5) \cdot (5 \times 10^4)$ *in standard form*

   ANSWER 2×10^{10}

2. *Work out* $(1.2 \times 10^7) : (4 \times 10^3)$

   ANSWER 3×10^3

3. *Write 0.(45) as a fraction in lowest terms*

   ANSWER $\frac{5}{11}$

4. *Write 0.1(6) as a fraction*

   ANSWER $\frac{1}{6}$ — multiply by 10 and by 100, then subtract.

5. *Without dividing, say whether* $\frac{9}{48}$ *terminates*

   ANSWER Cancel to $\frac{3}{16}$; $16 = 2^4$, so yes.

6. *The mass of a grain of sand is about* 6.6×10^{-5} *kg. Find the mass of* 2×10^6 *grains.*

   ANSWER $1.32 \times 10^2 = 132$ kg

7. *Is 0.101001000100001… rational? Explain.*

   ANSWER No — it never stops and never repeats, so it is irrational.

07 Homework

Homework — 6 problems

Algebra 8, pp. 62–67 (practical problems) and Cambridge Workbook 1.2.

1. *Write 0.000091 and 340 000 in standard form*
2. *Work out* $(5 \times 10^4) \cdot (6 \times 10^3)$ *in standard form*

3. *Work out* $(9 \times 10^8) : (3 \times 10^5)$

4. *Write* $\frac{7}{8}$ *and* $\frac{4}{9}$ *as decimals*

5. *Write* $0.(8)$ *as a fraction*

6. *Without dividing, decide whether* $\frac{11}{50}$ *terminates*

Control work 3 · Roots and rational exponents

The Chapter I assessment, followed by a work-on-mistakes lesson sorted by error type.

LESSON 36–37 | 2 × 40 MIN | ALGEBRA 8, §§8–10 | STAGE 9 · UNIT 1 CHECK

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess roots, rational exponents, standard form and decimals.
- Sort the errors by type rather than by question.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of §§8–10
Cambridge	Learner's Book pp. 9–20
Workbook	Workbook Unit 1

01 Explanation

Lesson 36 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each, 14 in total:

- tasks 1–2 · simplify a root (§8)
- task 3 · rationalise a denominator (§8)
- tasks 4–5 · evaluate a power with a rational exponent (§9)
- task 6 · simplify a letter expression (§10)
- task 7 · standard form or a recurring decimal (Cambridge insert)

Lesson 37 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. **Splitting a root over a sum** — writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.
2. **Numerator and denominator of the index swapped** — reading $8^{2/3}$ as $(8^3)^2$.
3. **A negative index treated as a negative answer** — $4^{-1/2}$ given as -2.
4. **Standard form not tidied** — leaving 12×10^9.

The Hard set below is seven pieces of wrong working built from these four.

02 Worked examples

EXAMPLE 1 **Find the error:** $\sqrt{16 + 9} = 4 + 3 = 7$

1. A root does not split over a sum.
   Only over a product or a quotient.

2. $16 + 9 = 25$
   Do the addition first.

3. $\sqrt{25} = 5$

ANSWER

5

EXAMPLE 2 **Find the error:** $4^{-1/2} = -2$

1. A negative index means a reciprocal, not a negative number.

2. $4^{-1/2} = \dfrac{1}{4^{1/2}}$

3. $= \dfrac{1}{2}$

ANSWER

$\dfrac{1}{2}$

03 Interactive model

Show each piece of wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: $\sqrt{16 + 9} = 7$

1. The product rule is $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$. There is no sum rule.

2. $16 + 9 = 25$, so the root is $\sqrt{25}$.

3. $= 5$, not 7. One counter-example is enough to kill the false rule.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Root	Ildiz	Корень
Exponent	Ko‘rsatkich	Показатель
Standard form	Standart ko‘rinish	Стандартный вид
Evaluate	Hisoblash	Вычислить
Justify	Asoslash	Обосновать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sqrt{25 + 144}$ equals:

- 17
- 13
- 169
- $5 + 12$

$8^{2/3}$ equals:

- 4
- 64
- 16
- 262144

$9^{-1/2}$ equals:

- -3
- $\frac{1}{3}$
- 3
- $-\frac{1}{3}$

20×10^9 in standard form is:

- 20×10^9
- 2×10^{10}
- 2×10^8
- 0.2×10^{11}

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Simplify* $\sqrt{48}$

   ANSWER $4\sqrt{3}$

2. ***Task 2.*** *Simplify* $\sqrt{6} \cdot \sqrt{24}$

   ANSWER 12

3. ***Task 3.*** *Rationalise* $\dfrac{4}{\sqrt{2}}$

   ANSWER $2\sqrt{2}$

4. ***Task 4.*** *Evaluate* $16^{1/2}$

   ANSWER 4

5. ***Task 5.*** *Evaluate* $27^{2/3}$

   ANSWER 9

6. ***Task 6.*** *Simplify* $a^{1/2} \cdot a^{3/2}$

   ANSWER a^2

7. ***Task 7.*** *Write 0.0035 in standard form*

   ANSWER 3.5×10^{-3}

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Simplify* $\sqrt{75}$

   ANSWER $5\sqrt{3}$

2. ***Task 2.*** *Simplify* $\sqrt{8} \cdot \sqrt{18}$

   ANSWER 12

3. ***Task 3.*** *Rationalise* $\dfrac{9}{\sqrt{3}}$

   ANSWER $3\sqrt{3}$

4. ***Task 4.*** *Evaluate* $64^{1/3}$

   ANSWER 4

5. ***Task 5.*** *Evaluate* $25^{-1/2}$

   ANSWER $\dfrac{1}{5}$

6. ***Task 6.*** *Simplify* $\dfrac{b^{5/4}}{b^{1/4}}$

   ANSWER b

7. ***Task 7.*** *Write* $\dfrac{2}{3}$ *as a decimal*

   ANSWER $0.(6)$

Hard · stretch and reason

7 PROBLEMS

1. *Find the error:* $\sqrt{16 + 9} = 7$

ANSWER Roots do not split over a sum. Correct: 5.

2. *Find the error:* $8^{2/3} = (8^3)^2$

ANSWER Index upside down. Correct: 4.

3. *Find the error:* $4^{-1/2} = -2$

ANSWER A negative index gives a reciprocal. Correct: $\frac{1}{2}$.

4. *Find the error:* $(4 \times 10^5)(5 \times 10^4) = 20 \times 10^9$

ANSWER Not tidied. Correct: 2×10^{10}.

5. *Find the error:* $\sqrt{a^2} = a$ *for every* a

ANSWER True only for $a \geq 0$; for $a < 0$ it is $-a$.

6. *Find the error:* $\frac{1}{\sqrt{2}} = \sqrt{2}$

ANSWER Rationalising gives $\frac{\sqrt{2}}{2}$, which is about 0.71, not 1.41.

7. *Find the error:* $0.(3) = \frac{3}{10}$

ANSWER $\frac{3}{10} = 0.3$ exactly, not recurring. Correct: $\frac{1}{3}$.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on, then these three.

1. *Simplify* $\sqrt{147}$ *and rationalise* $\dfrac{5}{\sqrt{5}}$
2. *Evaluate* $32^{3/5}$ *and* $49^{-1/2}$
3. Write out the four errors from this lesson in your own words with one example each.

Home / Grade 8 / Algebra / Quarter 2

Numerical inequalities

What “greater than” really means — one definition by subtraction that settles every comparison, however awkward the numbers.

LESSON 38–39 | 2 × 40 MIN | ALGEBRA 8, §11 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the definition of $a > b$ by means of the difference.
- Compare two numbers or expressions using the difference method.
- Read and write strict and non-strict inequalities.
- Place numbers correctly on a number line.

TEXTBOOK

National	§11, pp. 68–70
Cambridge	Learner’s Book pp. 96–102
Workbook	Workbook 4.3

01 Explanation

The definition

DEFINITION

$$a > b \text{ means } a - b > 0$$

$$a < b \text{ means } a - b < 0$$

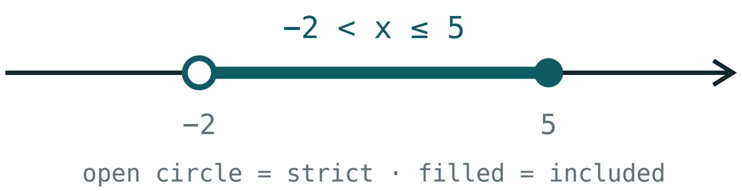
and $a = b$ means $a - b = 0$.

That is the whole of this lesson. Comparing two numbers is not a matter of looking at them — it is a matter of **subtracting and checking the sign**. For ordinary numbers this feels like overkill; for expressions with letters it is the only method that works.

On the number line, $a > b$ means a lies to the **right** of b.

Strict and non-strict

SYMBOL	READ AS	TYPE	ON THE LINE
$a > b$	a is greater than b	strict	open circle
$a < b$	a is less than b	strict	open circle
$a \geq b$	a is greater than or equal to b	non-strict	filled circle
$a \leq b$	a is less than or equal to b	non-strict	filled circle



An open circle excludes the boundary; a filled circle includes it.

Note that $5 \geq 5$ is **true** — “or equal to” only has to hold on one side.

The difference method in action

Compare $\frac{3}{7}$ and $\frac{4}{9}$ without decimals:

$$\frac{3}{7} - \frac{4}{9} = \frac{27 - 28}{63} = -\frac{1}{63} < 0$$

The difference is negative, so $\frac{3}{7} < \frac{4}{9}$. No decimals, no calculator, no rounding to argue about.

With letters it is the only route. To show that $a^2 + 1 > 2a$ for every a, subtract:

$$a^2 + 1 - 2a = (a - 1)^2 \geq 0$$

A square is never negative, so the difference is never negative — the inequality holds, with equality only when $a = 1$.

TWO USEFUL FACTS

> $a^2 \geq 0$ for every a, and $a^2 > 0$ unless $a = 0$. Almost every proof in this chapter ends by spotting a square.

02 Worked examples

EXAMPLE 1 **Compare $\frac{5}{8}$ and $\frac{7}{11}$**

1. $\frac{5}{8} - \frac{7}{11}$

   Subtract.

2. $= \frac{55 - 56}{88} = -\frac{1}{88}$

   Common denominator 88.

3. The difference is negative.

4. $\frac{5}{8} < \frac{7}{11}$

ANSWER

$\frac{5}{8} < \frac{7}{11}$

EXAMPLE 2 ***Prove that $a^2 + 4 \geq 4a$ for every a***

1. $a^2 + 4 - 4a$

   Take the difference.

2. $= a^2 - 4a + 4$

   Rearrange.

3. $= (a - 2)^2$

   A perfect square.

4. $(a - 2)^2 \geq 0$ always.

   So the difference is never negative.

ANSWER

True for every a, with equality at $a = 2$.

EXAMPLE 3 ***Which is greater: 2^{10} or 10^3?***

1. $2^{10} = 1024$
2. $10^3 = 1000$
3. $1024 - 1000 = 24 > 0$

   The difference is positive.

4. $2^{10} > 10^3$

ANSWER

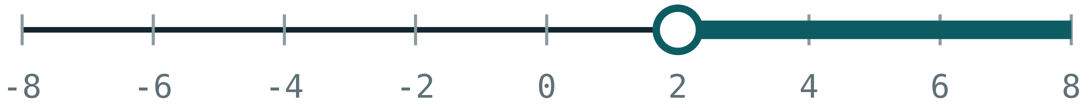
$2^{10} > 10^3$

03 Interactive model

Move the boundary and flip the symbol — the open and filled circles are the whole story.

Solutions on the number line

$$x > 2$$



inequality **x > 2** | boundary **2 (open)** | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inequality	Tengsizlik	Неравенство
Greater than	Katta	Больше
Less than	Kichik	Меньше
Strict inequality	Qat’iy tengsizlik	Строгое неравенство
Non-strict inequality	Qat’iy bo‘lmagan tengsizlik	Нестрогое неравенство
Difference	Ayirma	Разность
Positive	Musbat	Положительный
Negative	Manfiy	Отрицательный
Number line	Sonlar o‘qi	Числовая прямая
Compare	Taqqoslash	Сравнить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$a > b$ means:

- $a - b > 0$
- $b - a > 0$
- $a + b > 0$
- $ab > 0$

Which statement is true?

- $5 > 5$
- $5 \geq 5$
- $5 < 5$
- none of them

a^2 is:

- always positive
- never negative
- always negative
- sometimes negative

On the number line, a filled circle means:

- the value is excluded
- the value is included
- the inequality is strict
- nothing in particular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Compare 7 and 4 using the difference*

   ANSWER $7 - 4 = 3 > 0$, so $7 > 4$

2. *Compare −3 and −5*

   ANSWER $-3 - (-5) = 2 > 0$, so $-3 > -5$

3. *Is $6 \geq 6$ true?*

   ANSWER Yes.

4. *Is $6 > 6$ true?*

   ANSWER No.

5. *Write “x is at least 5” as an inequality*

   ANSWER $x \geq 5$

6. *Write “x is less than −2” as an inequality*

   ANSWER $x < -2$

7. *Compare* $\frac{1}{2}$ *and* $\frac{2}{5}$

   ANSWER $\frac{1}{2} > \frac{2}{5}$

Medium · the standard question 7 PROBLEMS

1. *Compare* $\frac{3}{7}$ *and* $\frac{4}{9}$

   ANSWER $\frac{3}{7} < \frac{4}{9}$

2. *Compare* $\frac{5}{8}$ *and* $\frac{7}{11}$

   ANSWER $\frac{5}{8} < \frac{7}{11}$

3. *Which is greater:* 2^{10} *or* 10^3*?*

   ANSWER $2^{10} = 1024 > 1000$

4. *Compare* $3\sqrt{2}$ *and* 4

   ANSWER $3\sqrt{2} \approx 4.24 > 4$

5. *Prove* $a^2 + 1 \geq 2a$

   ANSWER The difference is $(a - 1)^2 \geq 0$.

6. *Prove* $a^2 + 4 \geq 4a$

   ANSWER The difference is $(a - 2)^2 \geq 0$.

7. *Compare* $-\frac{2}{3}$ *and* $-\frac{3}{4}$

   ANSWER $-\frac{2}{3} > -\frac{3}{4}$

Hard · stretch and reason

7 PROBLEMS

1. *Prove that* $a^2 + b^2 \geq 2ab$ *for all* a, b

ANSWER The difference is $(a - b)^2 \geq 0$.

2. *Prove that* $a + \frac{1}{a} \geq 2$ *for* $a > 0$

ANSWER The difference is $\frac{(a-1)^2}{a} \geq 0$ because $a > 0$.

3. *Compare* $\frac{a}{b}$ *and* $\frac{a+1}{b+1}$ *for* $a > b > 0$

ANSWER The difference is $\frac{a-b}{b(b+1)} > 0$, so $\frac{a}{b} > \frac{a+1}{b+1}$.

4. *Prove that* $(a + b)^2 \geq 4ab$

ANSWER The difference is $(a - b)^2 \geq 0$.

5. *Which is greater:* $\sqrt{10}$ *or* 3.2?

ANSWER $10 < 10.24 = 3.2^2$, so $\sqrt{10} < 3.2$

6. *Show that* $a^2 + b^2 + 2 \geq 2a + 2b$

ANSWER The difference is $(a - 1)^2 + (b - 1)^2 \geq 0$.

7. *For which* a *is* $a^2 > a$?

ANSWER $a^2 - a = a(a - 1) > 0$ when $a < 0$ or $a > 1$.

07 Homework

Homework — 5 problems

Algebra 8, §11, pp. 68–70. Show the difference in every comparison.

1. *Compare* $\frac{4}{9}$ *and* $\frac{5}{11}$ *by the difference method*

2. *Compare* $-\dfrac{3}{5}$ *and* $-\dfrac{5}{8}$

3. *Prove that* $a^2 + 9 \geq 6a$ *for every* a

4. *Which is greater:* 3^5 *or* 5^3*?*

5. *Write as inequalities: "x is at most 7" and "y is greater than −1"*

Basic properties of numerical inequalities

What you may do to both sides — and the one operation that turns the sign around.

LESSON 40–41 | 2 × 40 MIN | ALGEBRA 8, §12 | STAGE 9 · 4.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State the four basic properties of inequalities.
- Multiply or divide by a negative number and reverse the sign.
- Use transitivity to chain comparisons.
- Decide which operations preserve an inequality and which do not.

TEXTBOOK

National	§12, pp. 71–74
Cambridge	Learner's Book pp. 96–102
Workbook	Workbook 4.3

01 Explanation

The four properties

FOR ANY NUMBERS A, B, C

1. **Transitivity.** If $a > b$ and $b > c$ then $a > c$.
2. **Adding.** If $a > b$ then $a + c > b + c$ — for **any** c, positive or negative.
3. **Multiplying by a positive number.** If $a > b$ and $c > 0$ then $ac > bc$.
4. **Multiplying by a negative number.** If $a > b$ and $c < 0$ then $ac < bc$ — **the sign reverses**.

Property 2 is why you may move a term from one side to the other, changing its sign, exactly as in an equation. Property 4 is the one thing an inequality does that an equation does not.

Why the sign reverses

Take $3 > 2$, which is certainly true, and multiply both sides by -1:

$$-3 \text{ and } -2$$

On the number line -3 sits to the **left** of -2, so $-3 < -2$. The order has turned round. Multiplying by a negative number reflects both numbers through 0, and reflection swaps left and right.

> THE RULE THAT COSTS THE MOST MARKS ALL YEAR
>
> From $-2x > 6$ you get $x < -3$, not $x > -3$. Dividing by -2 reverses the sign. Write the new sign **first**, then do the arithmetic.

Two consequences worth knowing

- **Reciprocals.** If $a > b > 0$ then $\frac{1}{a} < \frac{1}{b}$. Bigger denominator, smaller fraction.
- **Squares.** If $a > b > 0$ then $a^2 > b^2$. But this needs both numbers positive: $-5 > -7$ while $25 < 49$.

> A QUICK TEST TO BUILD THE HABIT
>
> Before applying any step, ask: “am I multiplying or dividing by something negative?” If yes, the sign turns. If no, it stays.

02 Worked examples

EXAMPLE 1 *Given $a > b$, compare $3a - 5$ and $3b - 5$*

1. $a > b$

   Given.

2. $3a > 3b$

   Multiply by 3, which is positive — sign unchanged.

3. $3a - 5 > 3b - 5$

   Subtract 5 from both sides — sign unchanged.

ANSWER

$3a - 5 > 3b - 5$

EXAMPLE 2 *Given $a > b$, compare $-2a$ and $-2b$*

1. $a > b$

   Given.

2. Multiplying by -2, a negative number.

   Property 4 applies.

3. $-2a < -2b$

   The sign reverses.

ANSWER

$-2a < -2b$

EXAMPLE 3 ***Given $2 < x < 5$, find the range of $3 - 2x$***

1. $2 < x < 5$

   Given.

2. $-4 > -2x > -10$

   Multiply throughout by -2 — both signs reverse.

3. $-10 < -2x < -4$

   Rewrite the smaller number on the left.

4. $-7 < 3 - 2x < -1$

   Add 3 throughout — signs unchanged.

ANSWER

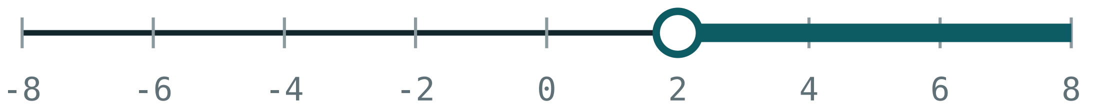
$-7 < 3 - 2x < -1$

03 Interactive model

Use the flip button to show that the same boundary gives two different solution sets.

Which side, and is the end included?

$$x > 2$$



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Property of an inequality	Tengsizlik xossasi	Свойство неравенства
Transitivity	Tranzitivlik	Транзитивность
Reverse the sign	Ishorani almashtirish	Изменить знак на противоположный
Both sides	Ikkala tomon	Обе части
Multiply by	Ko‘paytirish	Умножить на
Divide by	Bo‘lish	Разделить на
Preserve	Saqlanadi	Сохраняется
Equivalent inequality	Teng kuchli tengsizlik	Равносильное неравенство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

From $a > b$ it follows that:

- $a + 5 > b + 5$
- $a + 5 < b + 5$
- $a - 5 < b - 5$
- nothing

From $a > b$ and $c < 0$ it follows that:

- $ac > bc$
- $ac < bc$
- $ac = bc$
- nothing

$-3x > 12$ gives:

- $x > -4$
- $x < -4$
- $x > 4$
- $x < 4$

If $a > b > 0$ then:

- $\frac{1}{a} > \frac{1}{b}$
- $\frac{1}{a} < \frac{1}{b}$
- $\frac{1}{a} = \frac{1}{b}$
- it depends

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $a > b$. Compare $a + 4$ and $b + 4$

   ANSWER $a + 4 > b + 4$

2. $a > b$. Compare $a - 7$ and $b - 7$

   ANSWER $a - 7 > b - 7$

3. $a > b$. Compare $5a$ and $5b$

   ANSWER $5a > 5b$

4. $a > b$. Compare $-a$ and $-b$

   ANSWER $-a < -b$

5. Solve $-x > 3$

   ANSWER $x < -3$

6. Solve $2x > 8$

   ANSWER $x > 4$

7. Solve $-2x \geq 10$

   ANSWER $x \leq -5$

Medium · the standard question

7 PROBLEMS

1. $a > b$. Compare $3a - 5$ and $3b - 5$

   ANSWER $3a - 5 > 3b - 5$

2. $a > b$. Compare $-2a + 1$ and $-2b + 1$

   ANSWER $-2a + 1 < -2b + 1$

3. $a > b > 0$. Compare $\frac{1}{a}$ and $\frac{1}{b}$

   ANSWER $\frac{1}{a} < \frac{1}{b}$

4. $a > b > 0$. Compare a^2 and b^2

   ANSWER $a^2 > b^2$

5. Solve $-4x + 3 < 11$

   ANSWER $x > -2$

6. $2 < x < 5$. Find the range of $2x + 1$

   ANSWER $5 < 2x + 1 < 11$

7. Solve $\frac{x}{-3} > 2$

   ANSWER $x < -6$

Hard · stretch and reason

7 PROBLEMS

1. $2 < x < 5$. Find the range of $3 - 2x$

   ANSWER $-7 < 3 - 2x < -1$

2. $-1 < x < 4$. Find the range of $5 - x$

   ANSWER $1 < 5 - x < 6$

3. $a > b$. Is it always true that $a^2 > b^2$? Give a counter-example.

   ANSWER No — $-5 > -7$ but $25 < 49$. It needs $a > b > 0$.

4. $a > b$ and $c > d$. Prove $a + c > b + d$

   ANSWER $a + c > b + c$ and $b + c > b + d$, then transitivity.

5. $3 < x < 7$ and $1 < y < 2$. Find the range of $x - y$

   ANSWER $1 < x - y < 6$

6. $a > b > 0$. Compare $\frac{a}{b}$ and 1

   ANSWER $\frac{a}{b} > 1$ — divide $a > b$ by the positive number b.

7. Solve $5 - 3x \le 2x - 10$

   ANSWER $15 \le 5x$, so $x \ge 3$

07 Homework

Homework — 6 problems

Algebra 8, §12, pp. 71–74. Say out loud whether the sign turns before you write each line.

1. $a > b$. Compare $4a + 3$ and $4b + 3$
2. $a > b$. Compare $-5a$ and $-5b$
3. Solve $-3x + 2 > 14$
4. $1 < x < 6$. Find the range of $4 - x$

5. *$a > b > 0$. Compare* $\frac{2}{a}$ *and* $\frac{2}{b}$

6. *Give a counter-example to “$a > b$ implies $a^2 > b^2$”*

Adding and multiplying inequalities

Two inequalities pointing the same way can be added — and, when everything is positive, multiplied.

LESSON 42–43 | 2 × 40 MIN | ALGEBRA 8, §13 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Add two inequalities of the same direction.
- Multiply two inequalities of the same direction when all terms are positive.
- Estimate the range of a sum, difference, product or quotient.
- Explain why inequalities may not be subtracted or divided term by term.

TEXTBOOK

National	§13, pp. 75–79
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Adding

RULE

If $a > b$ and $c > d$ then $a + c > b + d$.

Inequalities pointing the **same way** may be added term by term. The proof is two steps of the last lesson: $a + c > b + c$ (add c) and $b + c > b + d$ (add b), then transitivity.

YOU MAY NOT SUBTRACT THEM

From $5 > 3$ and $10 > 1$ subtracting term by term would give $-5 > 2$, which is false. To handle a difference, reverse the second inequality first and then add: $c > d$ becomes $-c < -d$.

Multiplying

RULE

If $a > b > 0$ and $c > d > 0$ then $ac > bd$.

All four numbers must be **positive**. Drop that condition and the rule collapses: $2 > -5$ and $3 > -4$, but 6 is not greater than 20.

Nor may you divide term by term — invert the second inequality and multiply instead.

Estimating a range

This is what the rules are for. Given $3 < a < 5$ and $2 < b < 4$:

EXPRESSION	WORKING	RANGE
$a + b$	add the two ranges	$5 < a + b < 9$
$a - b$	reverse b: $-4 < -b < -2$, then add	$-1 < a - b < 3$
ab	multiply (all positive)	$6 < ab < 20$
$\frac{a}{b}$	reverse b: $\frac{1}{4} < \frac{1}{b} < \frac{1}{2}$, then multiply	$\frac{3}{4} < \frac{a}{b} < \frac{5}{2}$

The pattern for a difference or a quotient is always the same: **turn the second range round first**, then use the rule you are allowed to use.

02 Worked examples

EXAMPLE 1 ***Given $3 < a < 5$ and $2 < b < 4$, find the range of $a + b$***

1. $3 < a < 5$

   First range.

2. $2 < b < 4$

   Second range, same direction.

3. $3 + 2 < a + b < 5 + 4$

   Add the ends.

4. $5 < a + b < 9$

ANSWER

$5 < a + b < 9$

EXAMPLE 2 ***Given $3 < a < 5$ and $2 < b < 4$, find the range of $a - b$***

1. $2 < b < 4$

   Start from b.

2. $-4 < -b < -2$

   Multiply by -1 — both signs reverse.

3. $3 - 4 < a - b < 5 - 2$

   Now add.

4. $-1 < a - b < 3$

ANSWER

$-1 < a - b < 3$

EXAMPLE 3 ***Given $2 < x < 6$ and $1 < y < 3$, find the range of $\frac{x}{y}$***

1. $1 < y < 3$
2. $\frac{1}{3} < \frac{1}{y} < 1$

   Take reciprocals — the direction reverses.
3. $2 \cdot \frac{1}{3} < \frac{x}{y} < 6 \cdot 1$

   Multiply, all terms positive.
4. $\frac{2}{3} < \frac{x}{y} < 6$

ANSWER

$\frac{2}{3} < \frac{x}{y} < 6$

03 Interactive model

Set two boundaries and let the class predict the range of the sum before you show it.

Estimating a range

$3 < a < 5$ and $2 < b < 4$

1. Both point the same way, so they may simply be added.

2. $3 + 2 = 5$

and $5 + 4 = 9$

Add matching ends.

3. $5 < a + b < 9$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Add inequalities	Tengsizliklarni qo‘shish	Сложение неравенств
Multiply inequalities	Tengsizliklarni ko‘paytirish	Умножение неравенств
Same direction	Bir xil yo‘nalish	Одинаковый знак
Estimate	Baholash	Оценить
Range	Oraliq	Промежуток
Lower bound	Quyi chegara	Нижняя граница
Upper bound	Yuqori chegara	Верхняя граница
Term by term	Hadma-had	Почленно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

From $a > b$ and $c > d$ it follows that:

- $a + c > b + d$
- $a - c > b - d$
- $ac > bd$
- nothing

Multiplying two inequalities term by term needs:

- nothing extra
- all terms positive
- all terms whole numbers
- the same direction only

If $2 < a < 6$ then $-a$ satisfies:

- $-2 < -a < -6$
- $-6 < -a < -2$
- $2 < -a < 6$
- $-6 < -a < 6$

If $1 < b < 4$ then $\frac{1}{b}$ satisfies:

- $\frac{1}{4} < \frac{1}{b} < 1$
- $1 < \frac{1}{b} < 4$
- $\frac{1}{4} < \frac{1}{b} < \frac{1}{4}$
- it cannot be found

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $1 < a < 3$ and $2 < b < 5$. Find the range of $a + b$

ANSWER $3 < a + b < 8$

2. $0 < a < 2$ and $1 < b < 4$. Find the range of $a + b$

ANSWER $1 < a + b < 6$

3. $2 < a < 4$. Find the range of $-a$

ANSWER $-4 < -a < -2$

4. $1 < a < 5$. Find the range of $2a$

ANSWER $2 < 2a < 10$

5. $1 < a < 3$ and $2 < b < 4$. Find the range of ab

ANSWER $2 < ab < 12$

6. $3 < a < 6$. Find the range of $a + 10$

ANSWER $13 < a + 10 < 16$

7. *May two inequalities of the same direction be added?*

ANSWER Yes.

Medium · the standard question

7 PROBLEMS

1. $3 < a < 5$ and $2 < b < 4$. Find the range of $a + b$

ANSWER $5 < a + b < 9$

2. $3 < a < 5$ and $2 < b < 4$. Find the range of $a - b$

ANSWER $-1 < a - b < 3$

3. $3 < a < 5$ and $2 < b < 4$. Find the range of ab

ANSWER $6 < ab < 20$

4. $1 < b < 4$. Find the range of $\frac{1}{b}$

ANSWER $\frac{1}{4} < \frac{1}{b} < 1$

5. $2 < x < 6$ and $1 < y < 3$. Find the range of $\frac{x}{y}$

ANSWER $\frac{2}{3} < \frac{x}{y} < 6$

6. Show that $5 > 3$ and $10 > 1$ cannot be subtracted term by term

ANSWER It would give $-5 > 2$, which is false.

7. $1 < a < 2$ and $3 < b < 5$. Find the range of $2a + b$

ANSWER $5 < 2a + b < 9$

Hard · stretch and reason

7 PROBLEMS

1. $2 < a < 4$ and $1 < b < 3$. Find the range of $a^2 - b$

ANSWER $4 < a^2 < 16$ and $-3 < -b < -1$, so $1 < a^2 - b < 15$

2. The sides of a rectangle satisfy $3 < a < 5$ and $4 < b < 6$. Estimate the perimeter.

ANSWER $14 < P < 22$

3. The same rectangle: estimate the area.

ANSWER $12 < S < 30$

4. Give a counter-example to "$a > b$ and $c > d$ implies $ac > bd$".

ANSWER $2 > -5$ and $3 > -4$, but $6 < 20$.

5. $1 < x < 3$. Find the range of $\frac{1}{x+1}$

ANSWER $2 < x + 1 < 4$, so $\frac{1}{4} < \frac{1}{x+1} < \frac{1}{2}$

6. $2 < a < 3$ and $4 < b < 5$. Find the range of $\frac{b}{a}$

ANSWER $\frac{1}{3} < \frac{1}{a} < \frac{1}{2}$, so $\frac{4}{3} < \frac{b}{a} < \frac{5}{2}$

7. $0 < a < 1$. Compare a^2 with a.

ANSWER $a^2 < a$ — multiplying $a < 1$ by the positive number a.

07 Homework

Homework — 5 problems

Algebra 8, §13, pp. 75–79. Reverse the second range before any difference or quotient.

1. $4 < a < 7$ and $1 < b < 3$. Find the range of $a + b$
2. $4 < a < 7$ and $1 < b < 3$. Find the range of $a - b$

3. *$4 < a < 7$ and $1 < b < 3$. Find the range of ab*
4. *$4 < a < 7$ and $1 < b < 3$. Find the range of* $\frac{a}{b}$
5. *A rectangle has $2 < a < 4$ and $5 < b < 7$. Estimate its perimeter and its area.*

Raising a numerical inequality to a power

Squaring, cubing and taking roots — safe when both sides are positive, and full of traps when they are not.

LESSON 44–46 | 3 × 40 MIN | ALGEBRA 8, §14 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Raise both sides of an inequality to a natural power correctly.
- Take roots of both sides of an inequality.
- Use these rules to estimate the range of a square or a root.
- Give counter-examples when the positivity condition is dropped.

TEXTBOOK

National	§14, pp. 80–84
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

The two rules

FOR POSITIVE NUMBERS

If $a > b > 0$ and n is a natural number, then

$$a^n > b^n \quad \text{and} \quad \sqrt[n]{a} > \sqrt[n]{b}$$

Both operations preserve the order — as long as both sides are positive. Squaring $5 > 3$ gives $25 > 9$ ✓; taking square roots of $16 > 9$ gives $4 > 3$ ✓.

What goes wrong without the condition

AN EVEN POWER DESTROYS THE ORDER

> $-2 > -5$ is true, but $(-2)^2 = 4$ and $(-5)^2 = 25$, so $4 < 25$. Squaring turned the inequality round.

The reason is that an even power throws away the sign. An **odd** power keeps it, so cubing is safe for all numbers: $-2 > -5$ gives $-8 > -125$ ✓.

OPERATION	SAFE WHEN	EXAMPLE OF FAILURE
square, 4th power…	both sides positive	$-2 > -5$ but $4 < 25$
cube, 5th power…	always	—
square root	both sides ≥ 0	root of a negative does not exist

Using the rules to estimate

Given $3 < a < 5$, all positive, so squaring is safe:

$$9 < a^2 < 25$$

But given $-3 < a < 2$ you cannot square the ends. Think about what a^2 actually does on that interval: it is 0 at $a = 0$ and largest at the end furthest from 0, which is -3. So $0 \le a^2 < 9$.

THE HABIT TO BUILD

Before squaring an interval, ask whether it contains 0. If it does, the smallest value of a^2 is 0 — not the square of either end.

02 Worked examples

EXAMPLE 1 *Given $2 < a < 6$, find the range of a^2*

1. Both ends are positive, so squaring keeps the order.
   Check this first.
2. $2^2 = 4$ and $6^2 = 36$
3. $4 < a^2 < 36$

ANSWER

$4 < a^2 < 36$

EXAMPLE 2 *Given $-4 < a < 3$, find the range of a^2*

1. The interval contains 0, so the ends may not simply be squared.
2. $a^2 = 0$ at $a = 0$
   That is the smallest value.
3. The end furthest from 0 is -4, giving 16.
   That is the largest — but not attained.
4. $0 \le a^2 < 16$
   Note the ≤ on the left.

ANSWER

$0 \le a^2 < 16$

EXAMPLE 3 ***Given $4 < a < 9$, find the range of $\sqrt{a}$***

1. Both ends positive, so the root keeps the order.
2. $\sqrt{4} = 2$ and $\sqrt{9} = 3$
3. $2 < \sqrt{a} < 3$

ANSWER

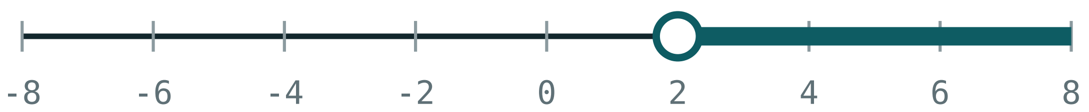
$2 < \sqrt{a} < 3$

03 Interactive model

Set the boundary either side of zero and ask what happens to the square of the interval.

Does the interval contain zero?

$$x > 2$$



inequality `x > 2`	boundary `2 (open)`	x = 0 works? **no**	x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Raise to a power	Darajaga ko‘tarish	Возведение в степень
Take the root	Ildiz chiqarish	Извлечь корень
Even power	Juft daraja	Чётная степень
Odd power	Toq daraja	Нечётная степень
Positive numbers	Musbat sonlar	Положительные числа
Condition	Shart	Условие
Counter-example	Qarshi misol	Контрпример
Monotonic	Monoton	Монотонный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

From $a > b > 0$ it follows that:

- $a^2 > b^2$
- $a^2 < b^2$
- $a^2 = b^2$
- nothing

$-2 > -5$. Squaring gives:

- $4 > 25$
- $4 < 25$
- $4 = 25$
- the same inequality

Cubing an inequality is safe:

- only for positives
- for all numbers
- never
- only for whole numbers

If $-3 < a < 2$ then a^2 satisfies:

- $9 < a^2 < 4$
- $0 \le a^2 < 9$
- $4 < a^2 < 9$
- $-9 < a^2 < 4$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *$5 > 3$. Square both sides.*

   ANSWER $25 > 9$

2. *$7 > 2$. Cube both sides.*

   ANSWER $343 > 8$

3. *$16 > 9$. Take square roots.*

   ANSWER $4 > 3$

4. *$1 < a < 4$. Find the range of a^2*

   ANSWER $1 < a^2 < 16$

5. *$9 < a < 25$. Find the range of $\sqrt{a}$*

   ANSWER $3 < \sqrt{a} < 5$

6. *Is squaring safe for $a > b > 0$?*

   ANSWER Yes.

7. *Is cubing safe for negative numbers?*

   ANSWER Yes — an odd power keeps the sign.

Medium · the standard question

7 PROBLEMS

1. $2 < a < 6$. Find the range of a^2

ANSWER $4 < a^2 < 36$

2. $4 < a < 9$. Find the range of $\sqrt{a}$

ANSWER $2 < \sqrt{a} < 3$

3. $1 < a < 2$. Find the range of a^3

ANSWER $1 < a^3 < 8$

4. $-5 < a < -2$. Find the range of a^2

ANSWER $4 < a^2 < 25$

5. $-5 < a < -2$. Find the range of a^3

ANSWER $-125 < a^3 < -8$

6. Show that $-2 > -5$ but $(-2)^2 < (-5)^2$

ANSWER $4 < 25$ — squaring reversed it.

7. $3 < a < 4$. Find the range of $\frac{1}{a^2}$

ANSWER $\frac{1}{16} < \frac{1}{a^2} < \frac{1}{9}$

Hard · stretch and reason

7 PROBLEMS

1. $-4 < a < 3$. Find the range of a^2

ANSWER $0 \le a^2 < 16$

2. $-1 < a < 5$. Find the range of a^2

ANSWER $0 \le a^2 < 25$

3. $2 < a < 3$. Find the range of $a^2 + a$

ANSWER $4 < a^2 < 9$, so $6 < a^2 + a < 12$

4. The side of a square satisfies $4 < a < 6$. Estimate its area and perimeter.

ANSWER $16 < S < 36$, $16 < P < 24$

5. Compare $\sqrt{5}$ and 2.3 without a calculator.

ANSWER $2.3^2 = 5.29 > 5$, so $\sqrt{5} < 2.3$

6. $0 < a < 1$. Compare a^3, a^2 and a.

ANSWER $a^3 < a^2 < a$ — each multiplication by $a < 1$ makes it smaller.

7. Given $1 < a < 4$, find the range of $\sqrt{a} + a$

ANSWER $1 < \sqrt{a} < 2$, so $2 < \sqrt{a} + a < 6$

07 Homework

Homework — 6 problems

Algebra 8, §14, pp. 80–84. Check for zero inside the interval before squaring.

1. $3 < a < 7$. Find the range of a^2
2. $1 < a < 8$. Find the range of $\sqrt[3]{a}$
3. $-6 < a < 2$. Find the range of a^2
4. $-3 < a < -1$. Find the range of a^3

5. *Compare* $\sqrt{7}$ *and 2.6 without a calculator*

6. *The side of a square satisfies* $5 < a < 8$. *Estimate its area.*

Control work 4 · Numerical inequalities

The Chapter II assessment on properties, adding, multiplying and powers — and the mistakes lesson that follows.

LESSON 47–48 | 2 × 40 MIN | ALGEBRA 8, §§11–14 | STAGE 9 · 4.3

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess the definition, the four properties, adding, multiplying and powers.
- Sort every lost mark into one of four error types.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of §§11–14
Cambridge	Learner's Book pp. 96–102
Workbook	Workbook 4.3

01 Explanation

Lesson 47 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · compare two numbers by the difference (§11)
- task 2 · a property applied to $a > b$ (§12)
- task 3 · solve a one-step inequality with a negative coefficient (§12)
- tasks 4–5 · estimate a sum, difference, product or quotient (§13)
- task 6 · estimate a square or a root (§14)
- task 7 · prove an inequality using a square (§11)

Lesson 48 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. **The sign not reversed** after dividing by a negative number.
2. **Inequalities subtracted or divided term by term.**
3. **An interval containing zero squared end to end**, losing the value 0.
4. **The positivity condition ignored** when multiplying or squaring.

The Hard set below is seven pieces of wrong working drawn from these four.

02 Worked examples

EXAMPLE 1 **Find the error: $-2x > 8$, so $x > -4$**

1. Dividing by -2 — a negative number.
   Property 4 applies.

2. The sign must reverse.

3. $x < -4$
   Check: $x = -5$ gives $10 > 8$ ✓

ANSWER

$x < -4$

EXAMPLE 2 **Find the error: $-3 < a < 2$, so $9 < a^2 < 4$**

1. The written range is impossible — 9 is not less than 4.
   That alone signals an error.

2. The interval contains 0, so the ends may not be squared directly.

3. $a^2 = 0$ at $a = 0$; the largest comes from -3.

4. $0 \le a^2 < 9$

ANSWER

$0 \le a^2 < 9$

03 Interactive model

Lesson 48: show each piece of wrong working, vote on the error type, then reveal.

Diagnose the error

Claimed: $-2x > 8 \Longrightarrow x > -4$

1. Dividing by

-2 is dividing by a negative number.

2. So the inequality sign turns:

$x <$ -4.

3. Test

$x =$ -5:

$-2(-5) = 10 > 8$ ✓, and

$-5 < -4$ ✓.

The test settles it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Estimate a range	Oraliqni baholash	Оценить промежуток
Reverse the sign	Ishorani almashtirish	Изменить знак неравенства
Counter-example	Qarshi misol	Контрпример
Prove	Isbotlash	Доказать
Condition	Shart	Условие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$-5x \geq 20$ gives:

- $x \geq -4$
- $x \leq -4$
- $x \geq 4$
- $x \leq 4$

From $a > b$ and $c > d$, which is guaranteed?

- $a - c > b - d$
- $a + c > b + d$
- $ac > bd$
- $\frac{a}{c} > \frac{b}{d}$

If $-2 < a < 5$ then:

- $4 < a^2 < 25$
- $0 \leq a^2 < 25$
- $-4 < a^2 < 25$
- $0 < a^2 < 4$

To prove $a^2 + 1 \geq 2a$ you should:

- try a = 3
- take the difference and factorise
- divide both sides by a
- square both sides

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Compare* $\frac{2}{5}$ *and* $\frac{3}{8}$ *by the difference*

   ANSWER $\frac{2}{5} > \frac{3}{8}$

2. ***Task 2.*** *$a > b$. Compare $a + 6$ and $b + 6$*

   ANSWER $a + 6 > b + 6$

3. ***Task 3.*** *Solve $-2x > 8$*

   ANSWER $x < -4$

4. ***Task 4.*** *$1 < a < 4$ and $2 < b < 5$. Find the range of $a + b$*

   ANSWER $3 < a + b < 9$

5. ***Task 5.*** *Same ranges: find the range of ab*

   ANSWER $2 < ab < 20$

6. ***Task 6.*** *$2 < a < 5$. Find the range of a^2*

   ANSWER $4 < a^2 < 25$

7. ***Task 7.*** *Prove $a^2 + 1 \geq 2a$*

   ANSWER The difference is $(a - 1)^2 \geq 0$.

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Compare* $\frac{4}{7}$ *and* $\frac{5}{9}$ *by the difference*

   ANSWER $\frac{4}{7} > \frac{5}{9}$

2. ***Task 2.*** *$a > b$. Compare $-3a$ and $-3b$*

   ANSWER $-3a < -3b$

3. ***Task 3.*** *Solve $-5x + 1 \le 11$*

   ANSWER $x \ge -2$

4. ***Task 4.*** *$2 < a < 5$ and $1 < b < 3$. Find the range of $a - b$*

   ANSWER $-1 < a - b < 4$

5. ***Task 5.*** *Same ranges: find the range of* $\frac{a}{b}$

   ANSWER $\frac{2}{3} < \frac{a}{b} < 5$

6. ***Task 6.*** *$9 < a < 36$. Find the range of $\sqrt{a}$*

   ANSWER $3 < \sqrt{a} < 6$

7. ***Task 7.*** *Prove $a^2 + b^2 \ge 2ab$*

   ANSWER The difference is $(a - b)^2 \ge 0$.

Hard · stretch and reason

7 PROBLEMS

1. *Find the error:* $-2x > 8$, so $x > -4$

   ANSWER The sign was not reversed. Correct: $x < -4$.

2. *Find the error:* from $5 > 3$ and $10 > 1$, $5 - 10 > 3 - 1$

   ANSWER Inequalities cannot be subtracted. Reverse the second, then add.

3. *Find the error:* $-3 < a < 2$, so $9 < a^2 < 4$

   ANSWER The interval contains 0. Correct: $0 \le a^2 < 9$.

4. *Find the error:* from $2 > -5$ and $3 > -4$, $6 > 20$

   ANSWER Multiplication needs all terms positive; here $6 < 20$.

5. *Find the error:* $a > b$, so $a^2 > b^2$

   ANSWER Needs $a > b > 0$. Counter-example $-2 > -5$ but $4 < 25$.

6. *Find the error:* $1 < b < 4$, so $1 < \frac{1}{b} < 4$

   ANSWER Reciprocals reverse the order. Correct: $\frac{1}{4} < \frac{1}{b} < 1$.

7. *Find the error:* "$a^2 \ge 2a$ is true for all a because $(a - 1)^2 \ge 0$"

   ANSWER The difference of a^2 and $2a$ is $a^2 - 2a = a(a - 2)$, not a square; the claim is false for $a = 1$.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quarter III opens with inequalities in one unknown.

1. Solve $-4x + 5 > 21$
2. $3 < a < 8$ and $2 < b < 4$. Find the ranges of $a + b$, $a - b$, ab and $\frac{a}{b}$

3. $-5 < a < 1$. *Find the range of* a^2

4. Write out the four errors from this lesson in your own words with an example each.